~/Akash Damle

// Backend Engineer

Badlapur, India · Open to remote roles

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$ summary

Backend engineer with 8+ years of total experience, building and operating systems in Django and Node/TypeScript. On client work I own the whole backend, sometimes as the only engineer and sometimes alongside a frontend developer, which means I have owned schema design, API surface, deployment and support rather than one slice of it.

Recent work has focused on making that ownership repeatable: typed API contracts generated from a single schema definition, integration test suites that run against a real database in CI, and infrastructure I can rebuild from scratch.

$ experience

Freelance Software Engineer — Independent · Remote

Mar 2021 – Present · 5 yr 8 mo

Django CRM and internal-tooling work for small organisations (schools, clinics and SMBs), delivered end to end and supported on an ongoing basis.

> Delivered 27+ client systems, each covering some combination of employee management, payroll calculation, attendance tracking, performance review and reporting.
> Designed the relational schema and REST API for each deployment, and built the admin and reporting surfaces on top.
> Provisioned and maintained self-managed Linux servers running PostgreSQL, including backups, TLS and upgrades. Hosting is part of the engagement rather than the client's problem.
> Worked directly with non-technical stakeholders to scope requirements, agree priorities and run acceptance.
> Also delivered custom websites with Django and Django CMS for clients whose needs stopped short of a full system.

stack: Django · Python · PostgreSQL · Django CMS · Linux · Nginx · Gunicorn

Junior Software Developer — Matalli Infotech · Dombivli

Aug 2018 – Feb 2021 · 2 yr 7 mo

First engineering role, on a small team maintaining internal web applications.

> Built and maintained internal applications on Django 2.2 LTS.
> Wrote unit and database-level tests to keep regressions out of releases.
> Implemented and worked with CI/CD pipelines to shorten the release cycle.
> Managed deployments across Linux servers, AWS and Heroku.
> Delivered production patches and bug fixes with minimal downtime.
> Learned project structure, code organisation and review practice in a team setting, and used Git throughout for collaboration.

stack: Django · Python · Git · Linux · AWS · Heroku · CI/CD

$ selected projects

Rootstock

2026

Write-up

A Django starter with accounts, settings and deploy files done, where every feature can be pruned in one verified commit.

> Twenty-three removable features and four presets (full, shared, api-only and minimal). Pruning is dry-run by default, needs a clean git tree, checks feature dependencies and is deliberately one-way.
> CI prunes every feature and preset and runs what remains. Doing that before the first push found seven defects, including a service worker that precached a stylesheet a prune had removed.
> Accounts use django-allauth with a UUID email user and required verification, as HTML pages and headless JSON. Hosted settings refuse to start without a secret key, allowed hosts, admin URL, database URL or email provider.
> A Django 5.2 LTS preset for shared hosting, with a Passenger upload bundle and a standard-library host probe that reports what a host can run before anything is installed.
> Every class and function has a docstring and no two are identical, enforced by a pre-commit check; a pre-push hook fails below 90 percent coverage.

stack: Python · Django · django-allauth · Django REST Framework · Django Ninja · OpenAPI · HTMX · Tailwind CSS · Celery · PostgreSQL · MySQL · Docker · Poetry · GitHub Actions · pytest

Pro Meet Desk

2026

Write-up

Django booking app: guests book without an account, PostgreSQL rules out double booking, and private notes never leave the professional.

> Double booking is impossible at the database: a btree_gist exclusion constraint, a per-professional advisory lock and threaded tests of simultaneous bookings and moves.
> Guests see only an area label until a booking is confirmed; the full address and meeting link arrive with the confirmation. A canary-string test checks that notes never reach any client-facing page, email or the admin.
> Client names, contact details, messages, addresses and notes are encrypted at rest with rotatable keys; clients are matched by a keyed hash of their email.
> An append-only audit trail, versioned consent, verified guest emails, and retention, export and erasure flows, at demo scope with the remaining legal work listed.
> Assistants can manage appointments and payments but never read notes; owner-only pages answer them with 403.

stack: Python · Django · PostgreSQL · HTMX · Tailwind CSS · cryptography · Brevo · Poetry · pytest

Lead Management Platform

2026

github.com/code-kasha/lead-platform · Live demo · Write-up

Django REST API: leads move through a validated pipeline with every change on a timeline, and the React app is typed from the API schema.

> The status pipeline (New, Contacted, Qualified, Proposal, Won or Lost) is enforced in one service function; a blocked change returns 400 with the reason.
> Every create, edit, status change, assignment and note is written to the timeline in the same transaction as the change.
> Admins manage every lead; members see and work on only the leads they created or are assigned to, and anything else returns 404.
> Access tokens last 15 minutes; refresh tokens rotate on every use and are blacklisted after use or on logout. The browser shares one refresh across every request waiting on it, because a rotated token works only once.
> 99 backend and 74 frontend tests. CI also runs flake8, a missing-migrations check, the stale-schema and stale-types check, and a Docker build with a smoke test of the running container.

stack: Python · Django · Django REST Framework · PostgreSQL · OpenAPI · React · TypeScript · TanStack Query · Docker · GitHub Actions · pytest · Vitest

Webhook Delivery

2026

github.com/code-kasha/webhook-delivery · Live demo · Write-up

Node.js and TypeScript service: signed webhooks with durable retries and a full attempt history, on PostgreSQL.

> Transactional fan-out, fenced worker leases and bounded retries retain events and attempt history across failures; delivery is at least once, with no ordering guarantee.
> 85 tests run against PostgreSQL in CI on Node 22 and 24, including concurrent claims, worker crashes, database outages, HTTPS verification and SSRF boundaries.
> Timestamped HMAC signatures, overlapping secret rotation and DNS/IP validation with address pinning protect the receiver and outbound request boundary.
> A controlled Render restart retained credentials, signing secrets, queued deliveries and prior attempt history; the retained webhook then received HTTP 204.

stack: Node.js · TypeScript · Fastify · PostgreSQL · Zod · OpenAPI · Docker · GitHub Actions · Vitest

Operations API

2026

github.com/code-kasha/operations-api · Live demo · Write-up

Django REST API: staff, shifts, leave, overtime and working-day payroll, where a finished pay run can never change.

> Four business roles (operations admin, HR, manager and employee) follow a documented permission matrix. Managers see their department and employees see themselves; anything else returns 404, and nobody reviews their own leave, timesheet or overtime.
> Payroll pro-rates base pay by working days for joiners, leavers, absences and unpaid leave, rounds half-up to the paisa, pays each approved overtime request exactly once and locks finished pay runs. It calculates gross pay only and claims no statutory compliance.
> Monthly attendance summary, payroll register and headcount reports reuse each module's role-scoped querysets, so a report never shows what its reader could not open.
> One command loads a fictional office with a month of history and a locked pay run, all or nothing. The public demo resets to it on every start, and the reset refuses to run anywhere else.
> 346 tests, also run once against PostgreSQL 17. CI checks lint, formatting, missing migrations and a stale OpenAPI schema, smoke-tests the Docker image, publishes releases from tags and redeploys the demo after each push.

stack: Python · Django · Django REST Framework · PostgreSQL · OpenAPI · Docker · GitHub Actions · pytest

bharat-post-dir

2023–2026

github.com/code-kasha/bharat-post-dir · Live demo · Write-up

India's 155,599 post offices as a lookup page, a JSON API and one download, each stating where the data came from.

> The lookup page costs one HTTP request and three database queries: inline CSS, no JavaScript, fonts or images. It works by keyboard and screen reader, in light and dark, and on phones.
> Imports are all-or-nothing. A short download is refused, identical rows merge, and offices listed more than once are kept and reported, never silently dropped.
> The whole-directory download is versioned by its SHA256 and sent with an ETag, so an unchanged directory is never downloaded twice.
> Anyone can load their own dataset. On a local clone an upload is temporary and private to one browser; in contributor mode it replaces the database and explains how to share it back. The hosted demo turns this off.
> CI deploys each push to main and waits until the demo's health check reports the new commit. 119 tests run against synthetic fixtures and never reach the network.

stack: Python · Django · Django REST Framework · SQLite · OpenAPI · Docker · GitHub Actions · pytest

CRM [In research and planning]

A CRM that arrives already set up for how your business works — sales, agency, real estate or field services — instead of making you configure it first.

$ skills

Backend	Django · DRF · Node.js · Express · Fastify · REST
Data	PostgreSQL · Constraints & locking · MongoDB · Redis · Query tuning
Quality	Vitest · Pytest · Zod · OpenAPI · Field encryption · Audit trails
AI tools	Claude Code · OpenAI Codex · ChatGPT
Languages	Python · TypeScript · JavaScript · SQL
Platform	Docker · CI/CD · Linux · AWS · Vercel
Frontend	React · Next.js · Tailwind · Redux · HTMX

$ education

B.Sc, Computer Science P V G's College of Science 2014 – 2018

$ training and certifications

Full Stack Development Internshala Oct 2025 – Apr 2026

Microservices with Node.js and React Udemy Apr 2024 – Sep 2025